

Tide retrospective: the multivariate sector bound via scaled sets

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

Fourth tide of the germbij arc, opening its multivariate phase. The three-tide 1D arc¹ landed the quantitative core of the germbij identifiability theorem in one dimension. The note’s actual Lemma 7.2 and Theorem 7.3 are d -dimensional, and the apparent obstacle to formalising them is the spherical-cap sector of the paper proof: polar coordinates and sphere measure are heavy Mathlib machinery. The idea of this tide is that they are unnecessary.

2 The thing formalised

Two statements in `Laplace/Multi/Sector.lean`, on ‘ $\iota \rightarrow \mathbb{R}$ ’ with the sup norm and Lebesgue measure, both sorry-free. The spherical cap is replaced by a *scaled set*: a measurable S of finite volume with $\|x\| \leq 2$ on S , whose windows are the pointwise dilates $u \cdot S$. The window lemma: if $c u^m \leq |a(ux)|$ and $K(ux) \leq 4C_0 u^2$ for $x \in S$, then

$$\text{vol}(S) u^d \cdot c^2 u^{2m} e^{-4C_0 t u^2} \leq \int_{u \cdot S} a^2 e^{-tK} dw, \quad d = \text{card } \iota,$$

by the Lebesgue scaling law $\text{vol}(u \cdot S) = u^d \text{vol}(S)$ and a constant lower bound. The Laplace-scale corollary (`sector_lower_bound_multi`): under $4 \leq r_0^2 t$, with $K \leq C_0 \|w\|^2$ for $\|w\| \leq r_0$ and the amplitude hypothesis at the single scale $u = (\sqrt{t})^{-1}$,

$$\text{vol}(S) \cdot c^2 e^{-4C_0} t^{-m-d/2} \leq \int_{(\sqrt{t})^{-1} \cdot S} a^2 e^{-tK} dw.$$

This is germbij Lemma 7.2 in $\mathbb{R}^d$, with the analytic input (a nonvanishing leading homogeneous part on a cap, controlling $|a(ux)| \geq c u^m$ on S for small u) factored into a hypothesis, exactly as the 1D tide factored $c w^m \leq |a|$. The deliberation confirmed the scaled-set model “faithfully captures the spherical-cap argument”. Numerical check: $d = 2$, $S = [1, 2]^2$, $a = w_1 w_2$, $K = w_1^2 + w_2^2$, $t = 100$ gives window integral 5.4×10^{-8} against bound 3.4×10^{-10} .

¹2026-08-08-19-11-tide-germbij-pencil,
2026-08-09-01-30-tide-germbij-identifiability.

2026-08-09-00-59-tide-germbij-sector,

3 Proof strategy

The window lemma is the 1D proof with the interval length replaced by the scaling law: measurability of the dilate,² the volume identity³ and the same pointwise product bound integrated by the constant-lower-bound lemma for set integrals. Points of the window are destructured by `rintro ⟨x, hxS, rfl⟩` from the pointwise-smul membership. The corollary re-runs the 1D substitution algebra with one new identity, $u^d = ((\sqrt{t})^d)^{-1}$, and the real-power split $t^{-m-d/2} = (t^m)^{-1} \cdot ((\sqrt{t})^d)^{-1}$.

4 Roadblocks and resolutions

Three small ones. The set-dilation notation ‘ $u \cdot S$ ’ is scoped: without `open Pointwise` the statement does not parse (instance-synthesis failures on `HSMul`). One `nlinarith` needed staging (first $\|x\|^2 \leq 4$, then the product hint with $C_0 u^2 \geq 0$). And as in the 1D tide, the linter caught an unused hypothesis: the window lemma does not need $0 \leq C_0$ once the K -bound is packaged, so the landed statement is again slightly stronger than the deliberated one.

5 What was Mathlib, what was new

Mathlib: the Haar scaling law, pointwise-set measurability, the set-integral constant bound, and the `rpow` API. New: the scaled-set reformulation itself, which is the tide’s real content: it converts the spherical-cap geometry of the paper proof into hypotheses that any concrete instantiation (a cap times a radial interval, a cube, a simplex) can discharge, and it makes the d -dimensional bound almost exactly as cheap as the 1D one (the two proofs differ by one lemma call and one `rpow` identity).

6 Lessons

When a paper proof reaches for polar coordinates to integrate over a sector, check whether the sector is only used through its volume scaling; if so, a scaled set plus `addHaar_smul` does the job with no sphere measure. And the recurring pair from this arc held again: try `exact` before `simp`, stage `nlinarith` hints, expect the linter to find a droppable hypothesis.

7 Follow-ups

- The measure-general pencil identity: generalise `pencil_identity_integrated` from Lebesgue on $\mathbb{R}$ to an arbitrary (σ -finite) measure, which its proof already supports; the multivariate composite needs it over ‘ $\iota \rightarrow \mathbb{R}$ ’.
- The multivariate composite (germbij Theorem 7.3 core in $\mathbb{R}^d$): the six-step assembly of the 1D identifiability tide, against `sector_lower_bound_multi`.
- A concrete instantiation lemma producing S, c, m from a nonzero homogeneous polynomial’s nonvanishing cap, connecting the factored hypotheses back to the analytic-germ setting of the note.

²`MeasurableSet.const_smul_of_ne_zero`.

³`Measure.addHaar_smul` with `Module.firank_pi`; the π volume is a Haar measure, so no Euclidean structure is needed.